C7 Organic Chemistry – Knowledge Organiser

AQA GCSE Chemistry • Set 1: Crude Oil and Alkanes **Paper 2**

#	Question	Answer
1	Crude oil	Finite mixture of hydrocarbons from ancient plankton
2	Hydrocarbon	Compound of hydrogen and carbon only
3	General formula of alkanes	C_nH_{2n+2}
4	First four alkanes	Methane, ethane, propane, butane
5	Fractional distillation separates by	Boiling point
6	Fractions from crude oil	LPG, petrol, kerosene, diesel, heavy fuel oil
7	Longer chains: boiling point	Higher
8	Longer chains: viscosity	Higher
9	Longer chains: flammability	Lower
10	Products of complete combustion	Carbon dioxide + water

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AQA GCSE Chemistry • Set 2: Cracking and Alkenes **Paper 2**

#	Question	Answer
1	Cracking	Breaking long hydrocarbons into shorter, more useful ones
2	Two methods of cracking	Catalytic cracking and steam cracking
3	Products of cracking	Shorter alkanes and alkenes
4	Why cracking is needed	Short chains are in higher demand
5	General formula of alkenes	C_nH_{2n}
6	Alkene functional group	C=C
7	Test for alkenes	Bromine water
8	Positive result for alkenes	Orange to colourless
9	Uses of alkenes	Making polymers and other chemicals
10	Unsaturated	Contains a C=C double bond

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AQA GCSE Chemistry • Set 3: Alkenes and Alcohols Paper 2 Triple only

#	Question	Answer
1	First four alkenes	Ethene, propene, butene, pentene
2	Alkene + hydrogen	Alkane (catalyst)
3	Alkene + steam	Alcohol (catalyst)
4	Alkene + bromine	Dibromoalkane
5	Alcohol functional group	–OH
6	First four alcohols	Methanol, ethanol, propanol, butanol
7	Alcohol + sodium	Hydrogen gas given off
8	Product of oxidising an alcohol	Carboxylic acid
9	Conditions for fermentation	Sugar solution, yeast, warm, no oxygen
10	Uses of ethanol	Fuel, solvent, drinks

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AQA GCSE Chemistry • Set 4: Carboxylic Acids, Esters and Polymers Paper 2 Triple only

#	Question	Answer
1	Carboxylic acid functional group	–COOH
2	First four carboxylic acids	Methanoic, ethanoic, propanoic, butanoic
3	Carboxylic acid + carbonate	Salt + water + carbon dioxide
4	Why carboxylic acids are weak acids	Partially ionise
5	Ester is made from	Carboxylic acid + alcohol (acid catalyst)
6	Ester from ethanol and ethanoic acid	Ethyl ethanoate
7	Monomers for addition polymerisation	Alkenes
8	Condensation polymerisation	Monomers with two functional groups join, losing a small molecule
9	Small molecule usually lost	Water
10	Monomers for a polyester	Diol + dicarboxylic acid

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AQA GCSE Chemistry • Set 5: Natural Polymers **Paper 2** **Triple only**

#	Question	Answer
1	Functional groups in an amino acid	Amine ($-NH_2$) and carboxylic acid (–COOH)
2	Amino acids polymerise to form	Polypeptides and proteins
3	DNA stands for	Deoxyribonucleic acid
4	Monomers of DNA	Nucleotides
5	Structure of DNA	Double helix
6	Examples of natural polymers	Proteins, starch, cellulose, DNA
7	Monomer of starch and cellulose	Glucose
8	Type of polymerisation in proteins	Condensation
9	Polymer from ethene	Poly(ethene)
10	Functional groups needed per monomer (condensation)	2